Hardware User Manual

AnyBus Series

BPFR/ BPHR

AJINEXTEK AXT

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Revision History

Manual	PCB	Comments
Rev. 1.0 issue 1.0	Rev. 1.0	2001. 12.18

1. Outline

1.1.Introduction

AnyBus Series product group is a module carrier board which supports *CompactPCI*® Bus, VME Bus, ISA Bus and PCI Bus. Among the AnyBus models, BPFR and BPHR are the basic mode carrier boards and they support PCI Full Size and PCI Half Size standards, respectively.

The basic mode carrier board is connected with the add-on AnyPack module which is composed of CPU module(AnyCPU series), motion module(AnyMotion series), digital I/O module(AnyDIO series), analog I/O module(AnyAIO series), distributed control communication module(AnyField series), etc. So various kinds of function boards can be implemented under any kind of control environment. Motion driving board, I/O control board and distributed control board can be composed depending on how you combine AnyPack modules on the AnyBus boards, and a combined board of more than two models can be also composed. If an independent CPU module(AnyCPU series) is loaded on the AnyBus board, it can reduce the load of a host CPU by taking charge of multi command processing, real time monitoring, real time motion control and I/O control in the CPU module, and the host CPU of the system can give a control command or monitor system conditions by communicating with the CPU module through DPRAM.

1.2.Application

1.2.1.Modules that can be Loaded

AnyPack is a common name of module that is loaded on the basic mode baseboard. AnyPack module into six groups by the function each module has. Those are AnyMotion series, a motor driving(motion control) module, Any DIO series, a digital input/output module, AnyAIO series, an analog input/output module, AnyCOM series, a universal communication module, AnyCPU series, an independent control CPU module and AnyField series, a distributed control master module.

The model name of AnyMotion module is fixed according to the motion chipset used and it is divided into 1axis, 2 axis and 4 axis modules by the number of motion axis. And it is also divided into 1U size module and 2U size module according to the module size. There are an input module, an output module and an input/output module in AnyDIO module, and up to 32 points I/O contacts are provided per each module. AnyAIO module has an input board and an output board which provide channels of various specifications. AnyField module and AnyCPU module provide various specifications according to the types of CPU used.

Modules that can be loaded among the AnyPack modules are shown below by series.

▼ AnyMotion series (Motion Module)

- ▶ SMC-1V01, SMC-1V02(*Active*[1)]) : 1 Axis motion control module with CAMC-5M/CAMC-FS (AjinExtek)(1U size)
- ▶ SMC-2V01, SMC-2V02 (*Active*) : 2 Axis motion control module with CAMC-5M/CAMC-FS (AjinExtek)(1U size)
- ▶ SMC-2V31 (*Preliminary*[2)]) : 2 Axis servo module with DSP(2U size)
- ▶ SMC-4V51 (*Preliminary*) : 4 Axis motion control module with MCX314 (NOVA) (2U size)
- ▶ SMC-2V54 (*Active*) : 2 Axis motion control module with MCX312 (NOVA) (1U size)
- ▶ SMC-2V53 (*Active*) : 2 Axis servo module with MC2120 (PMD)(1U size)
- ▶ SMC-4V52 (*Preliminary*) : 4 Axis servo module with PCL6045(PulseMotor)(2U size)

▼ AnyDIO • AnyAIO series (I/O Module)

- ▶ SIO-DI32,SIO-DI16 (*Active*) : 32/16 points digital input module(1U size)
- ▶ SIO-DO32P,SIO-DO16T (*Active*) : 32/16 points digital output module(1U size)
- ▶ SIO-DB32P,SIO-DB16T (*Active*) : 32 pts.(16/16)/16 pts.(8/8) digital I/O module(1U size)

[1)] *Active* **: Currently manufacturing products (2001. 10)**

[2)] *Preliminary* **: Products are currently scheduled to develop or under development (2001. 10)**

- SIO-AI4R (*Active*) : 4-ch analog input module(1U size)
- SIO-AO4R (*Active*) : 4-ch analog output module(1U size)
- SIO-CTx (*Preliminary*) : Counter-timer module(1U size)

▼ AnyCOM series (Communication Module)

- COM-234R (*Preliminary*) : 4-ch RS-232 serial communication module(1U size)
- COM-484R (*Preliminary*) : Communication module for both 4-ch RS-485 and RS-422 (1U size)
- COM-GP1R (*Preliminary*) : GPIB(IEEE 488) Communication Module(1U size)
- COM-ETxR (*Preliminary*) : Ethernet Communication Module(1U size)

▼ AnyCPU series (CPU Module)

- CPU-860 (*Active*) : CPU module with MPC-860(1U size)
- CPU-RABBIT (*Preliminary*) : CPU module with Rabbit 2000(1U size)
- CPU-8051 (*Active*) : CPU module with 8051 series CPU(1U size)
- CPU-ARM7 (*Preliminary*) : CPU module with Arm Processor(1U size)

▼ AnyField series (Distributed Master Module)

- CPU-860D (*Active*) : Distributed master module with MPC-860(2U size)

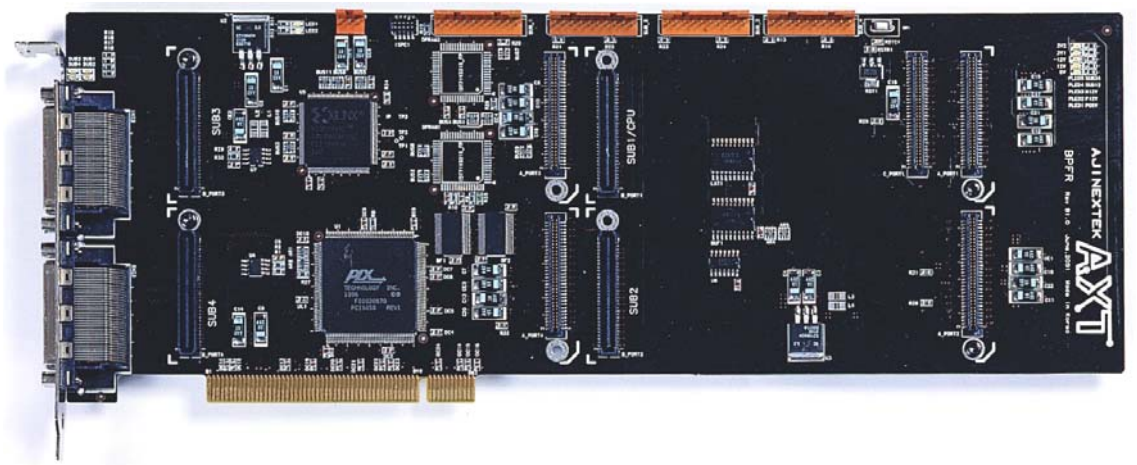
1.2.2.Uses

Combining AnyPack module products on the AnyBus series baseboard, it can be used widely from the PC based controller of a simple machine part to the complicated education purpose equipment, FA equipment and semiconductor manufacturing equipment.

- Event Counting
- Motion Control
- Motion Vision
- Robotics
- X-Y Table Control
- X-Y-Z Centering Control
- Loading/Unloading System
- Step Motor Control
- Machine Control
- Roll Feeding
- Pick & Placing
- R&D Use
- Industrial Digital I/O Control
- Signal Conversion
- Automatic Test Equipment
- Equipment Access
- Industrial Process Control (Regulation)
- Industrial Measurement and Monitoring
- Data Collecting of Multi-Channel
- Chemical Process Regulation
- Automatic Factory Operation
- Data Acquisition of Sensor
- Laboratory Measurement
- Data Collecting
- Industrial Serial Communication
- Multi User System
- Modem and Printer Regulation

2. Features

2.1. Specifications



Picture 1. BPFR

Table 1. Specifications of BPFR Board

Item	BPFR
Form Factor	PCI full size(long card)
Conformance	PCI Local Bus Spec. Rev. 2.1(32bit/33㎒)
Data Bus Width	32bit data mode
Interrupt	Automatic setting
Internal Setting Register	1Kbyte
Internal Slot	Support module with 4 local buses
DPRAM	2Kword(Option)
External Connector	Two 2 layer 68-pins for module & four ports of USB or RJ45 for distributed IO (Option)
Status LEDs	Check 5 power divisions and SUB 1,2,3,4 status
Power Requirements	5V, ±12V
Others	Max. I/O voltage of 150 ㎃ per each external connector pin

BPFR is a basic mode baseboard which supports PCI Local Bus Rev. 2.1, and it has total of 4 SUB slots including an independent control CPU slot. AnyPack module can be loaded on each slot, and 5V slot is used for PCI slot access.



Picture 2. BPHR

Table 2. Specifications of BPHR Board

Item	BPHR
Form Factor	PCI half size(short card)
Conformance	PCI Local Bus Spec. Rev. 2.1(32bit/33㎒)
Data Bus Width	32bit data mode
Interrupt	Automatic setting
Internal Setting Register	1Kbyte
Internal Slot	Support module with 2 local buses
External Connector	One 2 layer 68-pin for module & two ports of USB or RJ45 for distributed IO (Option)
Status LEDs	Check 4 power divisions and SUB 1,2 status
Power Requirements	5V, ±12V
Others	Max. I/O voltage of 150 ㎃ per each external connector pin

BPHR is a basic mode baseboard which supports PCI Local Bus Rev. 2.1. It has two SUB slots, and AnyCPU module cannot be loaded onto it contrary to BPFR. AnyPack module can be loaded on each slot, and 5V slot is used for PCI slot access.

3. Installation

3.1.Hardware Installation

Check the product's components first and then install a board in good order. The product is composed of a BPFR or BPHR board, a separate AnyPack module, an optional cable and a terminal block composed by its use. If you buy the BPFR board and AnyField module, the CON-D product that is a board for DCOM port and the connecting cable will be provided optionally.

① Switch off the power of PC then plug it out.

② Select the PCI slot (for 5V) to inset.

③ Remove the slot's bracket to use the selected slot.

④ To discharge static electricity which may have remained on clothes or the body, contact the metal part of a PC case with hands.

⑤ Insert a board in the selected PCI slot, and screw down the board on the slot with a fixing screw.

⑥ Check if it is installed well with the naked eye.

⑦ Plug in the power of PC, turn on the power switch and then make the system operate.

⑧ Check if 5 power LEDs on the upper right part of the board are normally on, and then check the SUB status LED equivalent to the slot is on when the module is loaded.

⑨ Check if booting was completed well.

⑩ Check if new hardware is searched in Windows 98 or Windows 2000 when booting up.

Complete the hardware installation with reference to the following installation process.

Windows 98, Windows 2000

Insert an installation CD.

Execute the automatic execution file. Be sure to install because it is the process of installing a system file for driving a device driver. Leave out this process when you install it on the already installed PCI slot again.

If hardware installation asks installation information, select the location (cd\AXTinstl\AnyBus \BPxR\Win9x or cd\AXTinst\AnyBus\BPxR\Win2k) where the installation file(BCxR.inf) exists.

Complete the installation by rebooting up.

Windows NT

Insert an installation CD.

Execute the automatic execution file. Be sure to install because it is the process of installing a system file for driving a device driver. Leave out this process when you install it on the already installed PCI slot again.

Move to the location (cd\AXTinstl\AnyBus\BPxR\WinNT) where the hardware installation file(BCxR.inf) exists.

Because automatic installation is not available in Windows NT, select the BCxR.inf file and install with an installation menu by clicking the right button of a mouse.

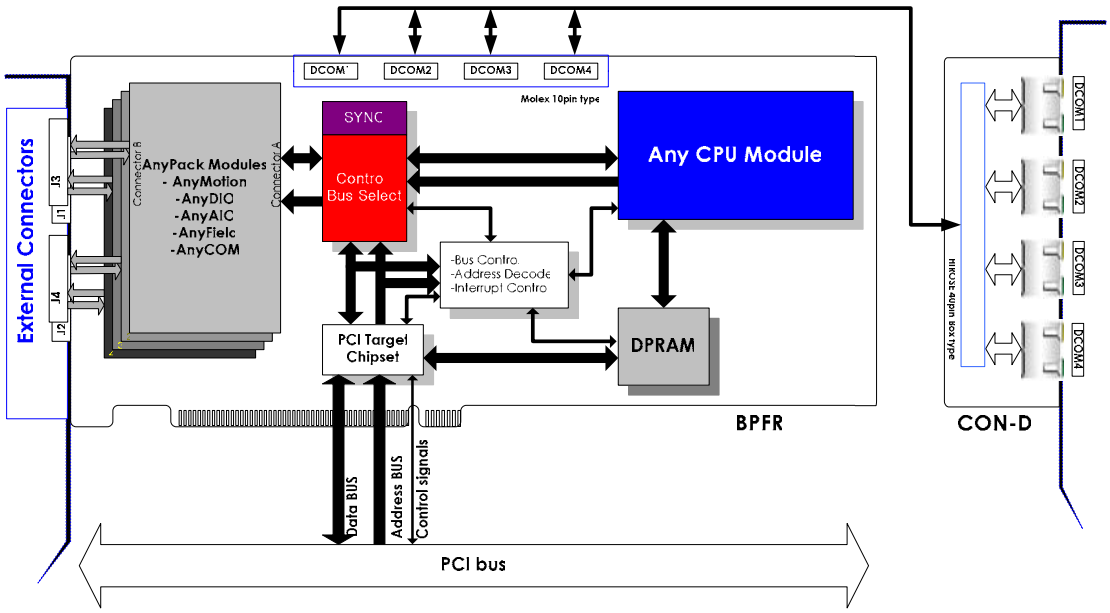
3.2.Software Installation

Install EzConfig referring to the EzConfig 1.0 user manual.

4. Configuration

4.1.Configuration of BPFR

The block diagram of BPFR board is as follows.



Picture 3. Block Diagram of BPFR Board

Table 3. Environment in Use of BPFR Board

Board Size	312×113. 5 ㎜
System Bus	PCI Local Bus Spec. Rev. 2.1(32bit/33㎒)
Necessary Space	1slot, 2 slots when connector board(CON-D) for distributed IO installed
Working Voltage	5V, ±12V
Connector	2 layer 68-pin CHAMP connector (J1, J2, J3, J4), CON-D : USB or RJ45 connector (DCOM1, DCOM2, DCOM3, DCOM4) (Option)

BPFR board is widely composed of PCI interface division, independent control CPU division, DPRAM division, SUB division, bus control and address decoder and external IO interface division.

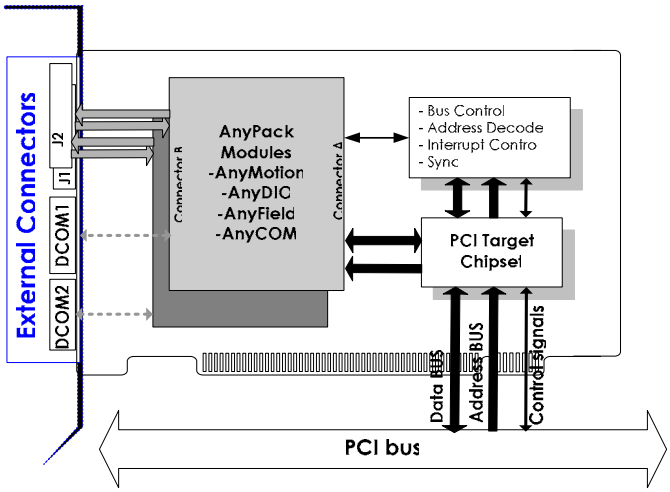
PCI interface division is perfectly compatible with PCI Local Bus Spec. Rev. 2.1 by using 9050 target chipset, and it is designed for 32bit data access. Local PCI that controls SUB and DPRAM applies 16bit data bus, and use a data bus of a module loaded on the SUB socket in 8 bits or 16 bits. Otherwise, a bus control division among the bus

control and address decoder divisions decides whether AnyPack module loaded on the SUB socket to use local PCI Bus or to operate by AnyCPU's control. As a local PCI Bus, an address decoder controls selection signal, writing signal, reading signal, data and address bus, etc. of each module when controlling SUB modules.

DPRAM is used only in case of using AnyCPU module, and it can be selected in 8bits or 16bits optionally according to the user demand. Namely, it can be communicated in 8bits (only Master loaded) or 16bits (Master/Slave loaded) between AnyCPU module and local PCI. SUB socket uses 64-pin PMC (PCI Mezzanine Card) connector, and AnyMotion, a motion control module, AnyDIO, a digital IO module, AnyAIO, an analog IO module, AnyField, a distributed master module, AnyCOM, a communication module and AnyCPU series, an independent control CPU module can be loaded.

The external IO interface division connected to the outside through the special or general purpose cable is composed of 68-pin CHAMP connector and DCOM Port which is a connector for a distributed master module, and each 68-pin connector and DCOM Port are connected with SUB modules. The 68-pin connector pin's signal varies according to the types of AnyPack module that is loaded on the SUB, and it can be divided into input, output and power. DCOM port is used when AnyField module is loaded, and a additional connector board (CON-D) is needed in order to connect with the external device(slave module). The CON-D board and connecting cable will be provided in case of purchasing AnyField module. For loading a CON-D board, another empty slot of a PC is needed.

4.2.Configuration of BPHR



Picture 4. Block Diagram of BPHR Board

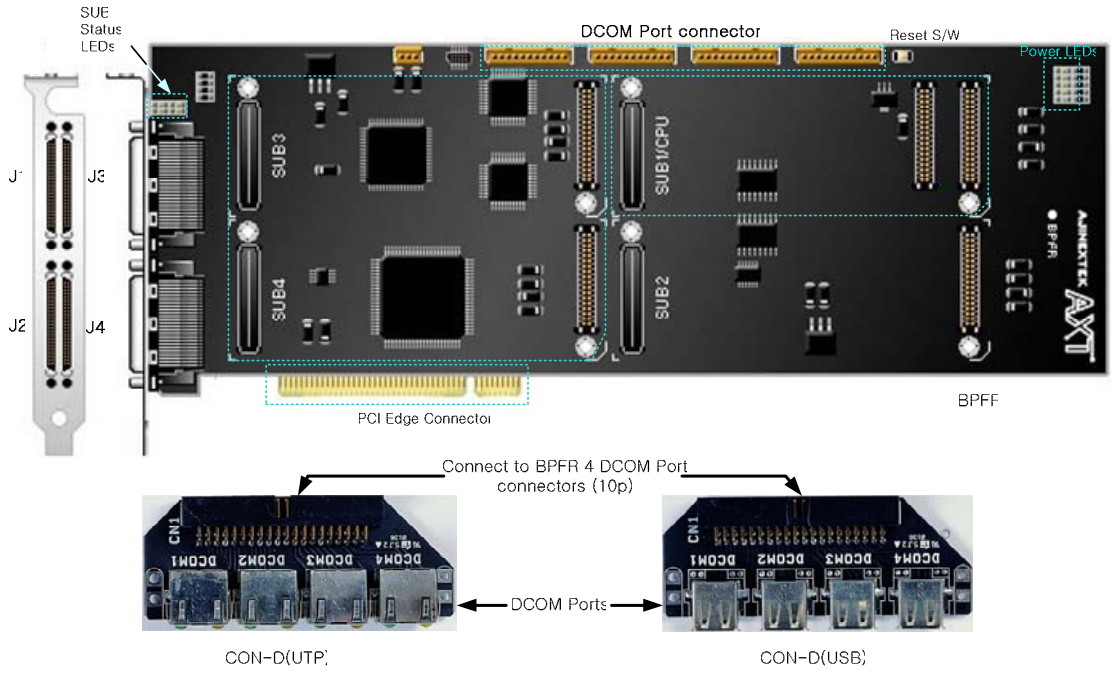
Table 4. Environment in Use of BPHR Board

Board Size	174.63×113.5 ㎜
System Bus	PCI Local Bus Spec. Rev. 2.1(32bit/33㎒)
Necessary Space	1slot
Working Voltage	5V, ±12V
Connector	Use 2 layer 68-pin CHAMP connector (J1,J2), USB or RJ45 connector(DCOM1, DCOM2)

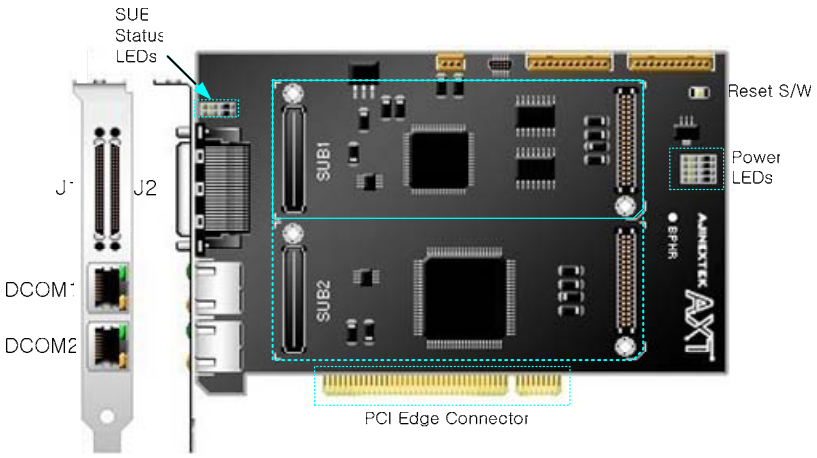
BPHR board is composed of PCI interface division, SUB division, address decoder and external IO interface division. PCI interface division is completely compatible with PCI Local Bus Spec. Rev. 2.1 using PCI9050 target chipset, and it is designed for 32bit data access. Local PCI can use 8bit or 16bit module with 16bit data bus. Address decoder controls each SUB module selection signal of local PCI Bus, write signal, read signal, data and address bus, etc. 64-pin PMC (PCI Mezzanine Card) typed SUB module connector can load with AnyMotion, AnyDIO, AnyAIO, AnyField and AnyCOM series. External IO interface division connected to the outside through the special or general purpose cable is composed of 68-pin CHAMP connector and DCOM Port which is a connector for the distributed master module. And each 68-pin connector pin's signal varies according to the types of AnyPack module that is loaded on the SUB, and it can be divided into input, output and power. DCOM Port is provided only when AnyField module is loaded. Since it provides UTP type and USB type separately, it is necessary to pay attention when order.

4.3.Hardware Description

4.3.1.Connector and LED



Picture 5. BPFR Board, CON-D Board and Connector



Picture 6. BPHR Board and Connector

BPFR, distributed IO connector CON-D and external connector, SUB module connector, indication LED of BPHR are displayed in the picture 5 and picture 6.

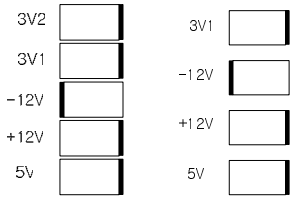
Each part is composed of SUB1,SUB2,SUB3,SUB4 slots for loading with the SUB module (BPHR has SUB1 and SUB2 only), power LEDs for making sure of the board's power state, connector for connecting the SUB module with an external device or terminal block through cable (J1,J2,J3,J4), distributed module connector (DCOM Ports), SUB status LEDs for making sure of the module loading state of each SUB slot and Reset Switch which turns SUB module's operation to the initial state. CON-D board connects with BPFR board using 40 pin cable, and the external connectors DCOM1,DCOM2,DCOM3,DCOM4 are connected with each module.

External device connectors J1,J2,J3,J4 are 68-pin CHAMP connectors, and they vary according to the types of module loaded on the SUB. The use of each pin varies according to which module is loaded. Each pin of the J1 connector is described in 4.7 by each module. DCOM Port, a distributed module connector, provides 8-pin RJ45 type and 4-pin USB type, so any type can be connected according to the interface method to be used in the distributed module. Use RJ-45 connector in case of connecting a distributed module in the UTP, RS-485, RS-422 or RS-232 mode, and USB connector in the USB mode. These two connectors are not exchangeable after shipping products. Therefore, distributed module users should make sure of the distributed control interface when placing an order.

Picture 5 and picture 6 show the figures with modules on the SUB slots. (Picture 5 shows 1U board loaded on SUB1/CPU and 2U board loaded over SUB3 and SUB4. The solid line of picture 6 shows 1U board loaded on SUB1, and the dotted line shows 2U board loaded over SUB1 and SUB2). When the components surface of BPFR and BPHR is placed on top, load the diagonally cut down parts (indicated parts) of the module board to be placed on the bottom right side. This part is indicated with silk on the board, so the module must be loaded as the shape it is indicated. AnyDIO, AnyAIO, AnyMotion, AnyCPU, AnyField and AnyCOM modules can be loaded on the SUB slots, but AnyCPU module can be loaded only on the SUB1/CPU slot. In case of using AnyCPU module, it communicates with a host CPU through DPRAM (option) and controls the rest 3 slots on the AnyCPU module. Therefore, each module's state and working sequence can be monitored on the host CPU.

The full module is divided into 1U board and 2U board. 1U board takes up one SUB slot and 2U SUB board takes up two. Among the modules on the BPFR board except AnyCPU module, 2U board can be loaded over SUB1-SUB2 and SUB3-SUB4. Each SUB slot is connected with the external connector J1,J2,J3,J4 and the distributed module connector DCOM1,DCOM2,DCOM3,DCOM4. SUB1 is connected with J1 and DCOM1, SUB2 is connected with J2 and DCOM2, SUB3 is connected with J3 and DCOM3 and SUB4 is connected with J4 and DCOM4. BPFR's DCOM Port is composed of 10 pin connector and it can be connected with the external device (distributed slave module) by connecting with a CON-D board through cable.

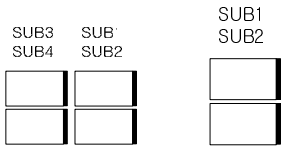
SUB status LED lights up the relevant LED when a module is loaded on the SUB slot. Power LED is for making sure of 3V power of SUB1 and SUB2, 3V power of SUB3 and SUB4 (None in BPHR), -12V power, +12V power and 5V power in order from above.



BPFR **BPHR**

3V2 marked LED on the BPFR board's silk shows the state of 3.3V power supplied to SUB1 and SUB2, and 3V1 shows the state of 3.3V power supplied to SUB3 and SUB4. -12V and +12V marked LEDs show the state of –12V and +12V power supply of SUB1, SUB2, SUB3, SUB4, respectively. 5V shows the state of 5V power supply, the master power.

BPHR board has no 3V2, and it only shows the state of 3V, -12V, +12V, 5V power supply.



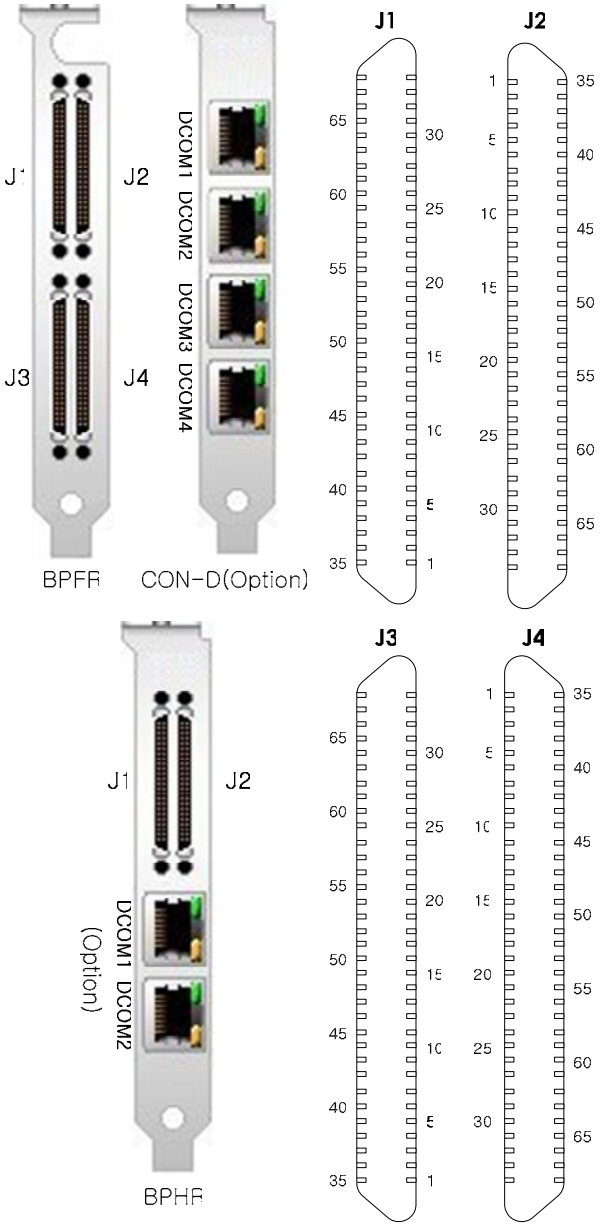
BPFR **BPHR**

LEDs marked on the upper left part of the BPFR and BPHR boards indicate whether the AnyPack module of the SUB module socket is loaded or not. SUB1, SUB2, SUB3 and SUB4 LEDs show the module loading state of SUB1, SUB2, SUB3 and SUB4, respectively.

4.4. Pin Array of I/O Connector

J Port(AnyMotion, AnyAIO, AnyDIO, AnyCOM External Connecting Terminal)

The external connectors, J1, J2, J3, J4 CHAMP connectors, have different meanings according to the module types. J1, J2, J3 and J4 are connected with the SUB1, SUB2, SUB3 and SUB4, respectively, and they are connected with the terminal blocks or the external devices directly through the cable.



J1, J2, J3, J4 Connector Signals

Motion Module
PULSE 출력신호 422 +/ -
DIR 출력신호 422+/-
아날로그 출력 신호
서보 드라이브 ON 신호
서보의 ALARM 신호
엔코더 A,B,Z상 입력의 422+/- 신호
범용 입력 신호 CH1,CH2
정/역방향 Over-run 방지 감속정지센서
정/역방향 Over-run 방지 급정지센서
급/감속정지 입력 신호
INPOSITION 출력신호
ALARM CLEAR 출력신호
잔여 펄스량 제거용 출력신호
범용 출력 신호 CH1,CH2
원점 검출용 센서 입력 신호
특정 위치에서 트리거 신호 출력
외부 jog 정/역방향 제어 입력

AIO/DIO Module
아날로그 입력 4CH
아날로그 출력 4CH
디지털 출력 16/32CH
디지털 입력 16/32CH
디지털 입출력 16/16CH

Picture 7. CHAMP Connector of BPFR Board (J1,J2,J3,J4), CHAMP Connector of BPHR Board (J1,J2)

DCOM Port (Distributed Module Connecting Terminal)

The part marked DCOM port is used when AnyField series product is loaded, and it supports RS-485, UTP, USB communication modes according to the functions of AnyField product. UTP and RS-485 modes use RJ45 connector and USB mode uses USB connector to connect with the external devices. Table 5 and table 6 described the connector pin's signal when RJ45 connector and USB connector are used.

Table 5. RJ45 Connector Type DCOM Port



RJ45 connector type

No	UTP Signal	RS-422(4 Line Signal)	RS-485(2 Line Signal)	RS-232C
1	TX+	TX+	TX_RX+	
2	TX-	TX-	TX_RX-	
3	RX+	RX+	TX_RX+	
4				RxD
5				TxD
6	RX-	RX-	TX_RX-	
7				GND
8				
LED1(GREEN)	LINK	LINK	LINK	LINK
LED2(YELLOW)	BUSY	BUSY	BUSY	BUSY

Table 6. USB Connector Type DCOM Port

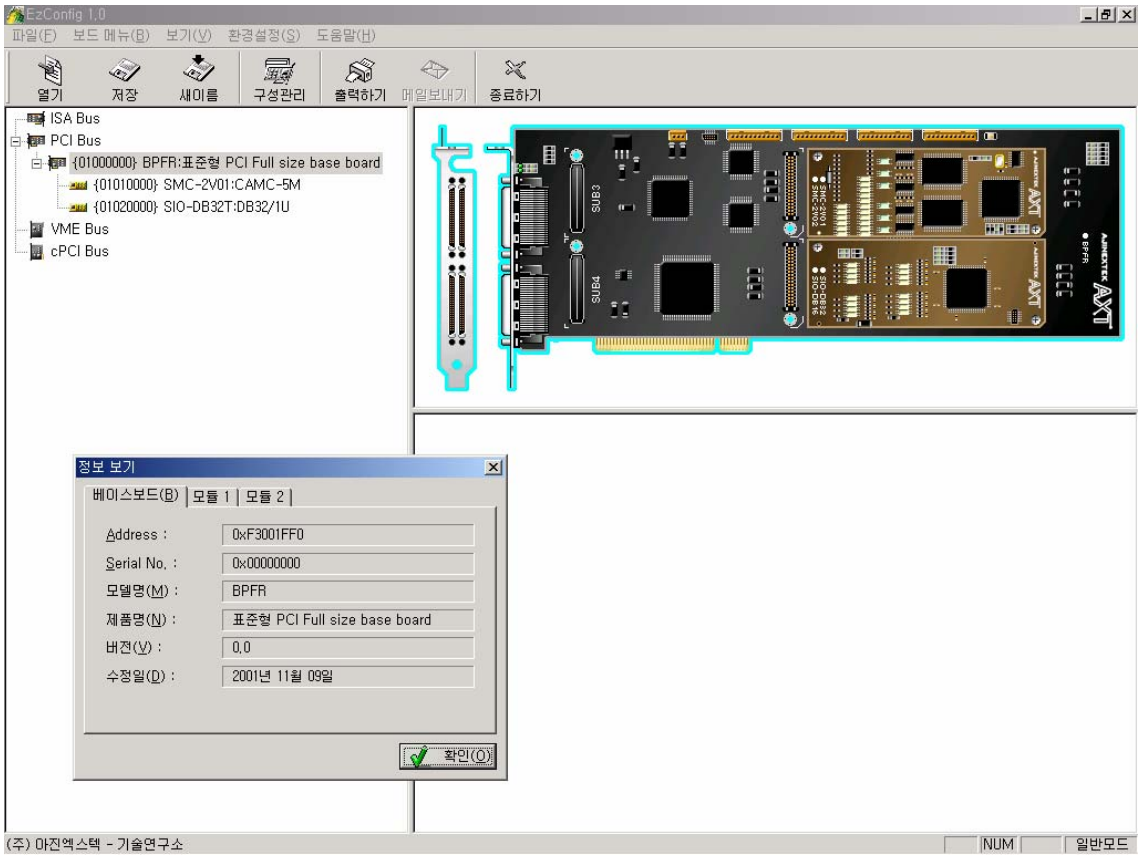


USB connector type

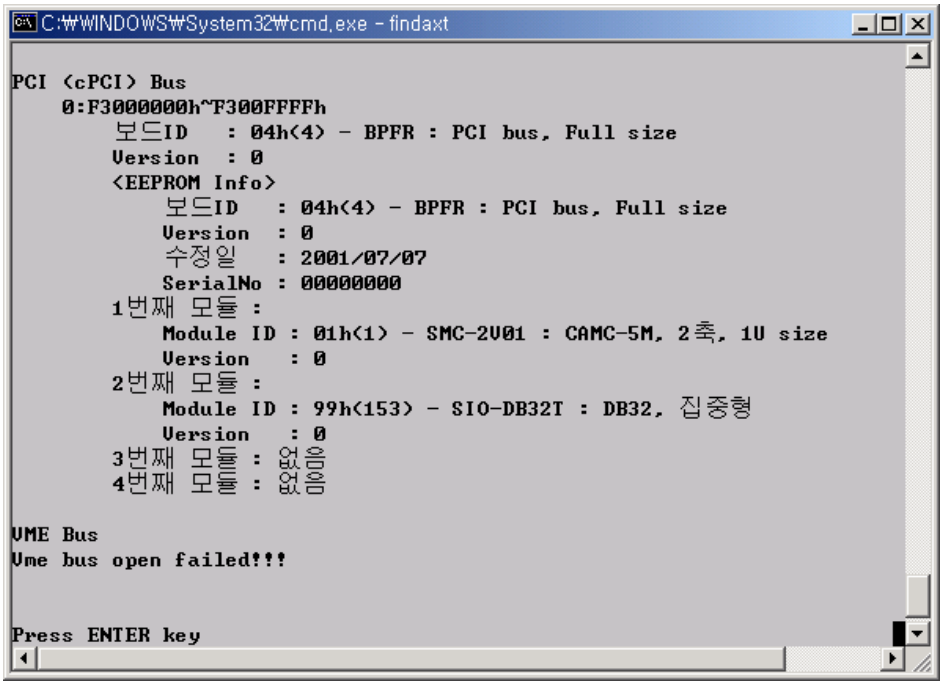
No	USB Signal
1	Power (+5V)
2	DATA -
3	DATA +
4	Power (0V)

4.5.Address Map Allocating Mode of the Board

The memory range allocated to each BPFR and BPHR is 64Kbyte. This range utilizes the automatically allocated range by PC's ROM BIOS, so it can be changed for each system used. If you use EzConfig program provided with the board, you can easily check the address used. Furthermore, address, slot location and utilizing IRQ of the board currently loaded on the system can be identified by using FindAXT.exe. Picture 8 is a screen that identifies a utilization address, shipping number of a board, a module loaded on the SUB slot in the BPFR system by using EzConfig. And each module's address loaded through an information view window can be identified. The following picture shows that it identifies information of the board that is loaded with SMC-2V01 module(SUB1) and SIO-DB32P module(SUB2) on the BPFR by the menu of the EzConfig board information view. You can get information about modules when you click each module's tab on the information view dialog window.



Picture 5. BPFR Information View Screen Using EzConfig Software



Picture 6. FindAXT Information View Screen

Picture 9 is a screen that makes sure of information of the board loaded on the PC by using FindAXT.exe. You can have information such as memory range in use, name of a board, revision date, manufacturing number, board type loaded on the SUB slot and so on by using FindAXT.exe.

4.6.Address Map of BPFR, BPHR Boards

4.6.1.Address Map

Table 7. Address Map of BPFR, BPHR Boards

A12(MSB) ~ A0(LSB)

Region	Offset	BPFR	BPHR	Remarks
		0000h – 1FFFh	0000h – 1FFFh	
0000h – 03FFh	—	Address Space Allocated to SUB1	Address Space Allocated to SUB1	1Kbyte
0400h – 07FFh	-	Address Space Allocated to SUB2	Address Space Allocated to SUB2	1Kbyte
0800h – 0BFFh	-	Address Space Allocated to SUB3	-	1Kbyte
0C00h – 0FFFh	-	Address Space Allocated to SUB4	-	1Kbyte
1000h – 17FFh	-	Base DPRAM	-	2Kbyte
1800h – 1FFFh	-	Range of Board Configuration	Range of Board Configuration	2Kbyte
	+000h	SYNC (Motion Module)	SYNC (Motion Module)	(1800h)
	+004h	Interrupt Enabled	Interrupt Enabled	(1804h)
	+006h	Interrupt Flag	Interrupt Flag	(1806h)
	+008h	Board LED	Board LED	(1808h)
	+3F0h	Preamble : 0xB6	Preamble : 0xB6	(1FF0h)
	+3F2h	Board ID(0x08)	Board ID(0x07)	(1FF2h)

Among the memory range allocated to BPFR, memory range in use is 8Kbyte in total and it is widely divided into 6. Those 6 ranges include four of 1Kbyte space allocated to each SUB, DPRAM 2Kbyte and 2Kbyte range to make sure of interrupt and board conditions. Read-write operations of 8 bits, 16 bits and 32 bits are possible for the whole memory range. In case of read-write operations in 8 bits, only even addresses have the effective value on the whole range. That is, odd addresses are not used.

A. SUB Allocated Space (4Kbyte – 0000h ~ 0FFFh)

SUB Base + Offset 0x00 : Preamble(Read Only)

0xB6 is read

SUB Base + Offset 0x02 : Module ID(Read Only)

SUB Base + Offset 0x04 ~ 0x3FF : Decided depending on modules (Refer to each manual)

Preamble is composed of continuous combination of '0' and '1', and the value of 0xB6 is always read. Each module type has its own ID, and the type of a board as well as its existence and nonexistence can be identified at each slot location using this ID and Preamble.

Table 8. AnyPack Module ID

Series	Model	ID
AnyMotion	SMC-2V01	01h
AnyMotion	SMC-1V01	03h
AnyMotion	SMC-2V31	1Fh
AnyMotion	SMC-4V52	34h
AnyMotion	SMC-2V54	36h
AnyDIO	SIO-DI32	97h
AnyDIO	SIO-DO32P	98h
AnyDIO	SIO-DB32P	99h
AnyAIO	SIO-AI4R	A1h
AnyCPU	CPU-860	C9h
AnyCPU	CPU-8051	CCh
AnyField	CPU-860D	CAh

Series	Model	ID
AnyMotion	SMC-2V02	02h
AnyMotion	SMC-1V02	04h
AnyMotion	SMC-4V51	33h
AnyMotion	SMC-2V53	35h
AnyDIO	SIO-DI16	97h
AnyDIO	SIO-DO16T	98h
AnyDIO	SIO-DB16T	99h
AnyAIO	SIO-AO4R	A2h
AnyCPU	CPU-RABBIT	CBh
AnyCPU	CPU-ARM7	CDh

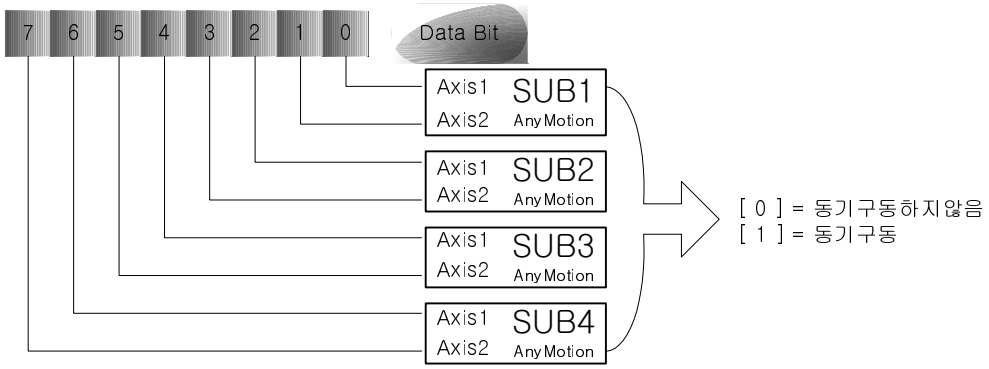
B. DPRAM Range (2Kbyte – 1000h ~ 17FFh)

Read and write are possible for the whole range, but DPRAM is optionally added only when AnyCPU module is loaded on the SUB1/CPU slot. When AnyCPU module that takes charge of independent control is loaded on the BPFR board's SUB1/CPU slot, AnyCPU module controls other SUB modules and it has done by communicating the module condition through DPRAM. It doesn't mean that CPCI bus controls the module directly. Interrupt is generated by doing read or write operation on the last byte allocated to the address map in DPRAM. That is, interrupt request signal toward the independent control CPU module is generated by reading 17FEh address or writing data. In the same manner with the independent control CPU module loaded on the SUB1/CPU slot, this process makes BPFR's interrupt generated by reading the last even address allocated to DPRAM or inputting data.

C. User Defined IO Range (2Kbyte – 1800h ~ 1FFFh)

① SYNC : Read and Write(1800h)

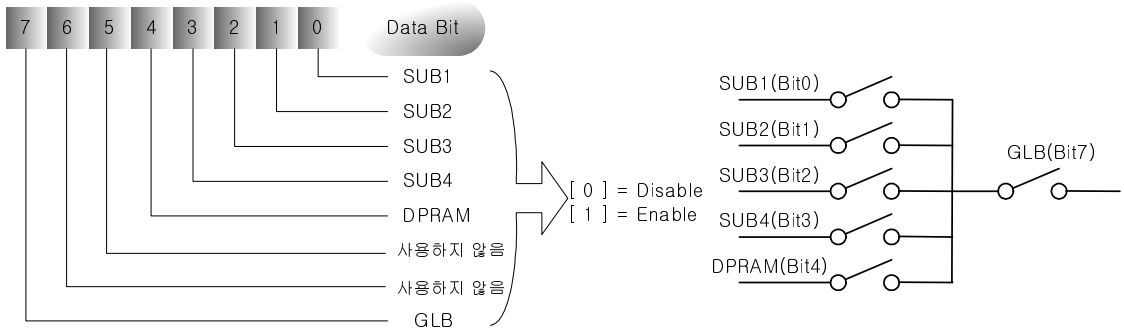
If AnyMotion module was loaded and there is necessity for driving motors at the same time, SYNC port makes simultaneous driving possible by activating each bit (into '1'). Only low-ranking 8 bits are used, not the high-ranking 8 bits.



For example, if both motors connected to SUB1's axis1 and SUB3's axis 2 must start driving simultaneously, a driving command must be given to the appropriated axes of SUB1 and SUB3 and the value of XX1XXXX1 must be used on the SYNC port in order to drive selected two axes simultaneously.

② Interrupt Enable Register : Read and Write(1804h)

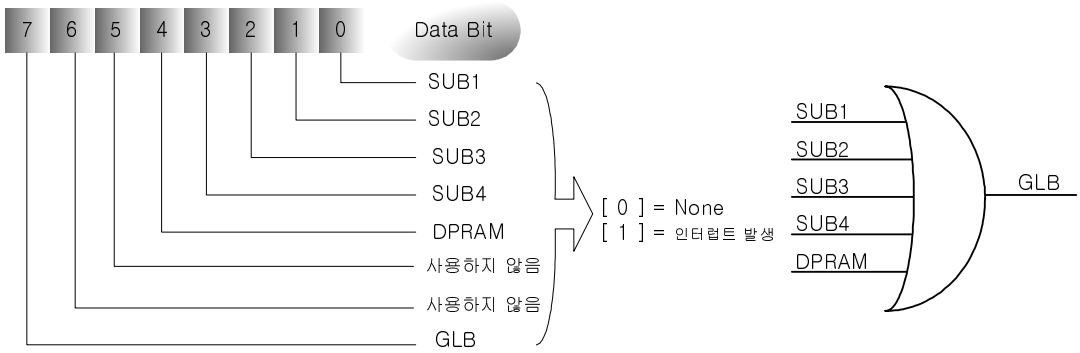
Interrupt enable register masks the interrupt request signal condition of SUB1, SUB2, SUB3, SUB4 and DPRAM, and makes it possible to use interrupt when the appropriate SUB bit is '1'.



Bit7's GLB is in the Global Enabled/Disabled state. If this bit becomes '0', it makes interrupt operation impossible regardless of the content of other bits. If it becomes '1', it makes interrupt request signal of DPRAM, SUB1, SUB2, SUB3 and SUB4 possible to process.

③ Interrupt Flag Register : Read Only(1806h)

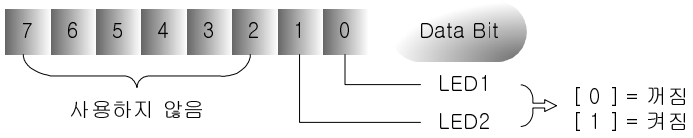
Interrupt flag register shows the interrupt request signal condition of DPRAM, SUB1, SUB2, SUB3 and SUB4. If each bit becomes '1', it shows that a module or DPRAM equivalent to the bit is in the interrupt request condition.



Bit7's GLB shows the interrupt request condition of a baseboard that becomes '1' if one bit out of the rest becomes '1'.

④ Board LED Register : Read and Write(1808h)

Board LED register for test is possible for read and write. It is used for the board test, and made to turn each board's LED on and off.



Inputting the value of 8 bits on this register, LED1 and LED2 shown in the picture 5 and picture 6 can be turned on and off according to the value of bit '0' and bit '1'. LED1 and LED2 are turned on when the equivalent bit becomes '1' and turned off when it becomes '0'.

⑤ Preamble : Read Only(1FF0h)

Preamble register is that a combination of '0' and '1' is arranged in order and the value of '10110110'(0xB6) is read. Since this value is always read on this register, the existence and nonexistence of the baseboard can be judged.

⑥ board id : Read Only(1FF2h)

As stated in the table 9, each baseboard has different board ID and it can be used for the automatic detection in software by reading this value and Preamble(0xB6).

Table 8. AnyBus Baseboard ID

	Model	ID		Model	ID
AnyBus	BPFR	04h	AnyBus	BPHR	03h
AnyBus	BIFR	02h	AnyBus	BIHR	01h
AnyBus	BV6R	06h	AnyBus	BV3R	05h
AnyBus	BC6R	08h	AnyBus	BC3R	07h

4.6.2.Optional Module

Table 9. Functional Combination When Optional Module is Loaded

( ) Means that CPU module is loaded.

Model Name	Platform	Size	AnyPack Model Supported				
			CPU	# of DIO Contacts	# of AIO Channels	# of Distributed Communication Ports	# of Motion Axes
BPFR	PCI	Full	1	128(96)	16(12)	4(Optional Slot)	8(6)
BPHR		Half	-	64	8	2	4
BIFR	ISA	Full	1	128(96)	16(12)	4(Optional Slot)	8(6)
BIHR		Half	-	64	8	2	4
BV6R	VME	6U	1	128(96)	16(12)	4	8(6)
BV3R		3U	-	64	8	2	4
BC6R	CPCI	6U	1	128(96)	16(12)	4	8(6)
BC3R		3U	-	64	8	2	4

In the table 10, the maximum number of DIO contacts, AIO contacts, distributed communication ports and motion axes is displayed which can be composed of BPFR with a CPU board and 3 or 4 SUB slots. The maximum composition list of BPHR with 2 SUB slots is displayed below, and the ways of composing with the other basic mode baseboards are also arranged.

If SUB1 slot is not used as a CPU slot, the number of IO contacts and driving motor axes is decided by AnyPack module loaded on each SUB slot. The number of maximum driving motor axes that can be composed of 2 axes of AnyMotion module per each slot is 8 in BPFR and 4 in BPHR. It is composed of four AnyDIOs with 32 contacts each, so it can be composed up to 128 points. Select input and output modules so that 128 points input board, input 64/output 64 points board and 128 points output board can be composed. And a small size system's controller can be settled with a board by accepting AnyMotion, AnyDIO or AnyAIO on a single baseboard. For example, a centering control system controller with 32 points digital output and 32 points digital input on the 2-axis motor

system can be composed by loading one SMC-2V02 and two SIO-DB32P on the BPFR baseboard. In case of using SUB1 as a CPU slot on the BPHR, 96 DIO contacts can be obtained by loading the maximum of 32 points AnyDIO on the three SUB slots. This constituent method can be changed into input 96 points (three SIO-DI32), output 96 points board (three SIO-DO32), input 48 points/output 48 points (three SIO-DB32) and so on depending on which AnyDIO board is used. When using a CPU slot, load with AnyAIO board so that it can have up to 12 AIO contacts as shown in the table.

Since AnyField module takes up the space of 2U, the BPFR can accommodate up to two AnyField modules and use four DCOM ports. The BPHR loads with one AnyField module and connects with the external distributed module through two DCOM ports.

4.7.Pin Array of CHAMP Connector (J1, J2, J3, J4) by Optional Module

4.7.1.Examples When AnyMotion Series Module is Loaded

When AnyMotion series optional module is loaded on the basic mode baseboard, the pin array of 68-pin CHAMP connector is as follows. AnyMotion module and AnyDIO module are described as an example in the table below. The item J1 is the case when a motion module is loaded on SUB1 and J2, J3, J4 are used when SUB2, SUB3, SUB4 are loaded, respectively.

e.g. When SMC-2V01, SMC-2V02 modules are loaded

Table 10. Description of 68-Pin CHAMP Connector Pin When SMC-2V01, 02 Modules are Loaded

J1	Pin Name	Dir.	Description	J1	Pin Name	Dir.	Description
1	P1_P24V	I	External +24V Power	35	P1_24VGND	I	External Power Supply
2	P1_PULS+	O	Pulse Output Signal (422+)	36	P1_PULS-	O	Pulse Output Signal (422-)
3	P1_DIR+	O	DIR Output Signal (422+)	37	P1_DIR-	I	DIR Output Signal (422-)
4	P1_SVON	O	Servo Drive ON Signal	38	P1_INP	O	Servo INPOSITION Signal
5	P1_ALM	I	Servo ALARM Signal	39	P1_ALMC	O	Servo ALARM CLEAR Signal
6	P1_LGND		LOGIC GND	40	P1_CLR	I	Servo Residual Pulse Clear Signal
7	P1_A+	I	Encoder A-Phase Input Signal (422+)	41	P1_A-	I	Encoder A-Phase Input Signal (422-)
8	P1_B+	I	Encoder B-Phase Input Signal (422+)	42	P1_B-	I	Encoder B-Phase Input Signal (422-)
9	P1_Z+	I	Encoder Z-Phase Input Signal (422+)	43	P1_Z-	O	Encoder Z-Phase Input Signal (422-)
10	P1_IN2	I	General Input Pin 2	44	P1_OUT2	O	General Output Pin 2
11	P1_IN3	I	General Input Pin 3	45	P1_OUT3	O	General Output Pin 3
12	N.C			46	N.C		
13	N.C			47	N.C		
14	P1_SSTOP	I	Slow Down Stop	48	P1_SLMT+	I	Positive Slow Down Limit
15	P1_SLMT-	I	Negative Slow Down Limit	49	P1_ELMT+	I	Positive End Limit
16	P1_ELMT-	I	Negative End Limit	50	P1_ORG	I	Origin Sensor Signal (General Input 0)
17	P1_EMGN	I	Emergency Stop (Emergency Stop Signal)	51	P1_TRIG	O	Trigger Occurrence Signal at the Special Location
18	P2_P24V	I	External +24V Power	52	P2_24GND	I	External Power Supply
19	P2_PULS+	O	Pulse Output Signal (422+)	53	P2_PULS-	O	Pulse Output Signal (422-)
20	P2_DIR+	O	DIR Output Signal (422+)	54	P2_DIR-	I	DIR Output Signal (422-)
21	P2_SVON	O	Servo Drive ON Signal	55	P2_INP	O	Servo INPOSITION Signal
22	P2_ALM	I	Servo ALARM Signal	56	P2_ALMC	O	Servo ALARM CLEAR Signal
23	P2_LGND		LOGIC GND	57	P2_CLR	I	Servo Residual Pulse Clear Signal
24	P2_A+	I	Encoder A-Phase Input Signal (422+)	58	P2_A-	I	Encoder A-Phase Input Signal (422-)
25	P2_B+	I	Encoder B-Phase Input Signal (422+)	59	P2_B-	I	Encoder B-Phase Input Signal (422-)
26	P2_Z+	I	Encoder Z-Phase Input Signal (422+)	60	P2_Z-	O	Encoder Z-Phase Input Signal (422-)
27	P2_IN2	I	General Input Pin 2	61	P2_OUT2	O	General Output Pin 2
28	P2_IN3	I	General Input Pin 3	62	P2_OUT3	O	General Output Pin 3
29	N.C			63	N.C		
30	N.C			64	N.C		
31	P2_SSTOP	I	Slow Down Stop	65	P2_SLMT+	I	Positive Slow Down Limit
32	P2_SLMT-	I	Negative Slow Down Limit	66	P2_ELM+	I	Positive End Limit

J1	Pin Name	Dir.	Description	J1	Pin Name	Dir.	Description
33	P2_ELMT-	I	Negative End Limit	67	P2_ORG	I	Origin Sensor Signal (General Input 0)
34	P2_EMGN	I	Emergency Stop(Emergency Stop Signal)	68	P2_TRIG	O	Trigger Occurrence Signal at the Special Location

If SMC-2V01, SMC-2V02 modules are loaded, the external connectors J1, J2, J3, J4 will be connected with a signal line necessary for 2-axis control per each connector. Only P1_ marked pins will be used if SMC-1V01, SMC-1V02 modules are loaded. For the name of each pin and the detailed connecting method with the servo system, refer to the board manual of each AnyMotion module [1].

4.7.2.Examples When AnyDIO Series Module is Loaded

e.g. When SIO-DB32P module is loaded

DB32T is a digital I/O board with 16 points of input and 16 points of output, and it is electrically isolated with the external power that is connected through a connector. As shown in the table below, pin numbers 10 and 44, used as signal lines in SMC-2V01 module, are used as external power in DB32T. So meanings of a power pin and each signal line used on each module vary according to the module loaded on the SUB. Therefore, be sure to refer to the AnyPack module when you connect the external devices and terminal blocks [2].

Table 11. Description of 68-Pin CHAMP Connector Pin When SIO-DB32P Module is Loaded

J1	Pin Name	Dir.	Description	J1	Pin Name	Dir.	Description
1	P12_P24V		+24V Power of INPORT1/OUTPORT1	35	P12_N24V		0V Power of INPORT1/OUTPORT1
2	P1_IN0	I	Inport 1, bit0	36	P2_OUT0	O	Outport 1, bit 0
3	P1_IN1	I	Inport 1, bit1	37	P2_OUT1	O	Outport 1, bit 1
4	P1_IN2	I	Inport 1, bit2	38	P2_OUT2	O	Outport 1, bit 2
5	P1_IN3	I	Inport 1, bit3	39	P2_OUT3	O	Outport 1, bit 3
6	P1_IN4	I	Inport 1, bit4	40	P2_OUT4	O	Outport 1, bit 4
7	P1_IN5	I	Inport 1, bit5	41	P2_OUT5	O	Outport 1, bit 5
8	P1_IN6	I	Inport 1, bit6	42	P2_OUT6	O	Outport 1, bit 6
9	P1_IN7	I	Inport 1, bit7	43	P2_OUT7	O	Outport 1, bit 7
10	P34_P24V	I	+24V Power of INPORT2/OUTPORT2	44	P34_N24V	O	0V Power of INPORT2/OUTPORT2
11	P3_IN0	I	Inport 2, bit8	45	P4_OUT0	O	Outport 2, bit 8
12	P3_IN1	I	Inport 2, bit9	46	P4_OUT1	O	Outport 2, bit 9

[1] **Caution** : Even though the same AnyMotion Series board is used, the use of f pin varies with each model loaded. So be sure to refer to the module manual.

[2] **Caution** : Refer to the module manual if different models of the same AnyDIO Series are loaded.

When you use the other AnyPack module in addition to the AnyMotion module and AnyDIO module described as examples, refer to each module's manual so that there wouldn't be any trouble in connection.

J1	Pin Name	Dir.	Description	J1	Pin Name	Dir.	Description
13	P3_IN2	I	Inport 2, bit10	47	P4_OUT2	O	Outport 2, bit 10
14	P3_IN3	I	Inport 2, bit11	48	P4_OUT3	O	Outport 2, bit 11
15	P3_IN4	I	Inport 2, bit12	49	P4_OUT4	O	Outport 2, bit 12
16	P3_IN5	I	Inport 2, bit13	50	P4_OUT5	O	Outport 2, bit 13
17	P3_IN6	I	Inport 2, bit14	51	P4_OUT6	O	Outport 2, bit 14
18	P3_IN7	I	Inport 2, bit15	52	P4_OUT7	O	Outport 2, bit 15
19	N.C			53	N.C		
20	N.C			54	N.C		
21	N.C			55	N.C		
22	N.C			56	N.C		
23	N.C			57	N.C		
24	N.C			58	N.C		
25	N.C			59	N.C		
26	N.C			60	N.C		
27	N.C			61	N.C		
28	N.C			62	N.C		
29	N.C			63	N.C		
30	N.C			64	N.C		
31	N.C			65	N.C		
32	N.C			66	N.C		
33	N.C			67	N.C		
34	N.C			68	N.C		

5. Accessories (Optional Items)

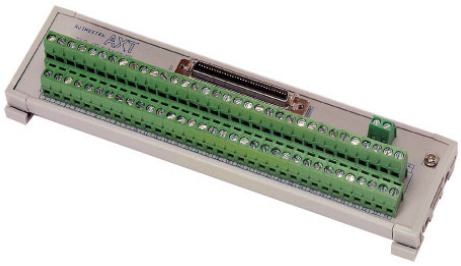
5.1.Cable



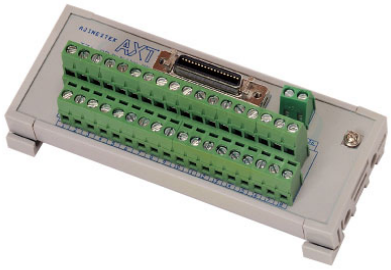
Picture 10. C68/36TS(Left) and C68(0.8-1.27)(Right)

In addition to the AnyPack, there are two types of cable, C68 (0.8-1.27) and C68/36TS, that are provided separately. These cables connect the external connector (J1, J2, J3, J4) of the AnyBus baseboard with the terminal block (see picture 11) in one to one. 68 pins compose twist pairs in numbers 1 and 35, 2 and 36, ….. , 34 and 68, and up to 150mA current can be flowed per each pin. C68/36TS and C68 (0.8-1.27) are used together with T68-PR and T36-PR, respectively. Each cable is 1m, 2m, 3m long in length, and it can be supplied up to 5m ling on the user's demand.

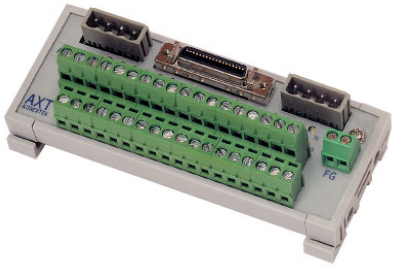
5.2.Terminal Block



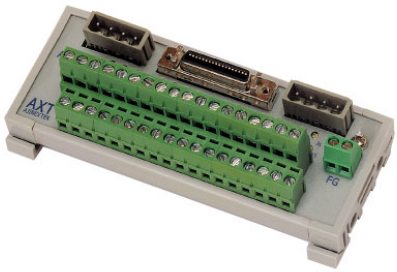
T68-PR One-To-One Terminal Block



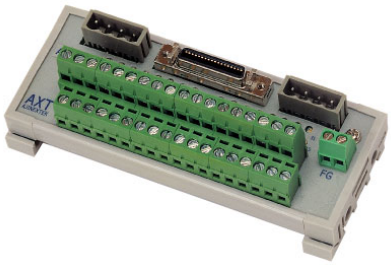
T36-PR One-To-One Terminal Block



APC-EI36 Digital Input Terminal Block



APC-EO36 Digital Output Terminal Block



APC-EB36 Digital I/O Terminal Block

Picture 11. Types of Terminal Block – One-To-One Terminal Block (Left) and High Quality I/O Terminal Block (Right)

A terminal block is provided separately on the AnyPack module and it varies according to the use of each module. 2-axis and 4-axis motor driving AnyMotion modules use T68-PR (one-to-one) and 1-axis motor driving AnyMotion module uses T68-PR or T36-PR (one-to-one) terminal block. In the AnyDIO module, a terminal block with a current amplification circuit (500mA per pin) or general terminal block connected one-to-one can be used on the

output stage. T68-PR terminal block is connected with the 68-pin CHAMP connector's pin in one-to-one in case of using C68pin (0.8-1.27) cable, and its pin number is same as the 68-pin CHAMP connector.

Terminal blocks such as T68-PR and T36-PR are designed to be able to load on the 35mm wide slide stationary stand, and pins that are connected with the devices can be fixed with a screw.

Table 12. Pin Relationship between Connector and Terminal Block

J1	T68-PR	T36-PR
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9
10	10	10
11	11	11
12	12	12
13	13	13
14	14	14
15	15	15
16	16	16
17	17	17

J1	T68-PR	T36-PR
18	18	18
19	19	
20	20	
21	21	
22	22	
23	23	
24	24	
25	25	
26	26	
27	27	
28	28	
29	29	
30	30	
31	31	
32	32	
33	33	
34	34	

J1	T68-PR	T36-PR
35	35	19
36	36	20
37	37	21
38	38	22
39	39	23
40	40	24
41	41	25
42	42	26
43	43	27
44	44	28
45	45	29
46	46	30
47	47	31
48	48	32
49	49	33
50	50	34
51	51	35

J1	T68-PR	T36-PR
52	52	36
53	53	
54	54	
55	55	
56	56	
57	57	
58	58	
59	59	
60	60	
61	61	
62	62	
63	63	
64	64	
65	65	
66	66	
67	67	
68	68	

5.3.Composition of Cable and Terminal Block by Each Module

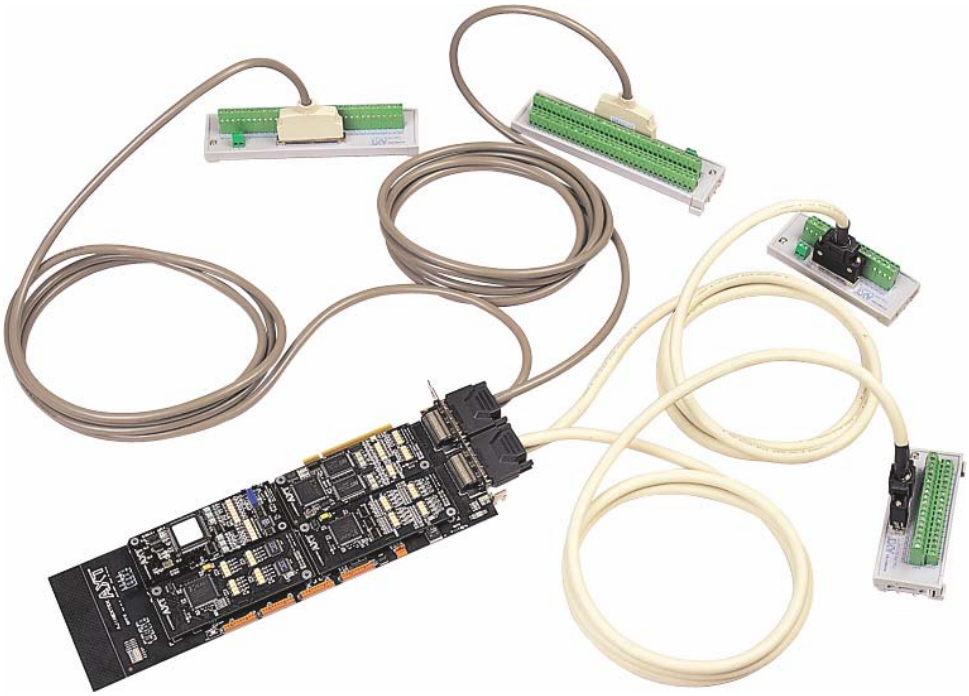
The cable and terminal block for connecting each AnyPack module with the external devices can be composed as the table below.

Table 13. Terminal Block and Cable by Each Module

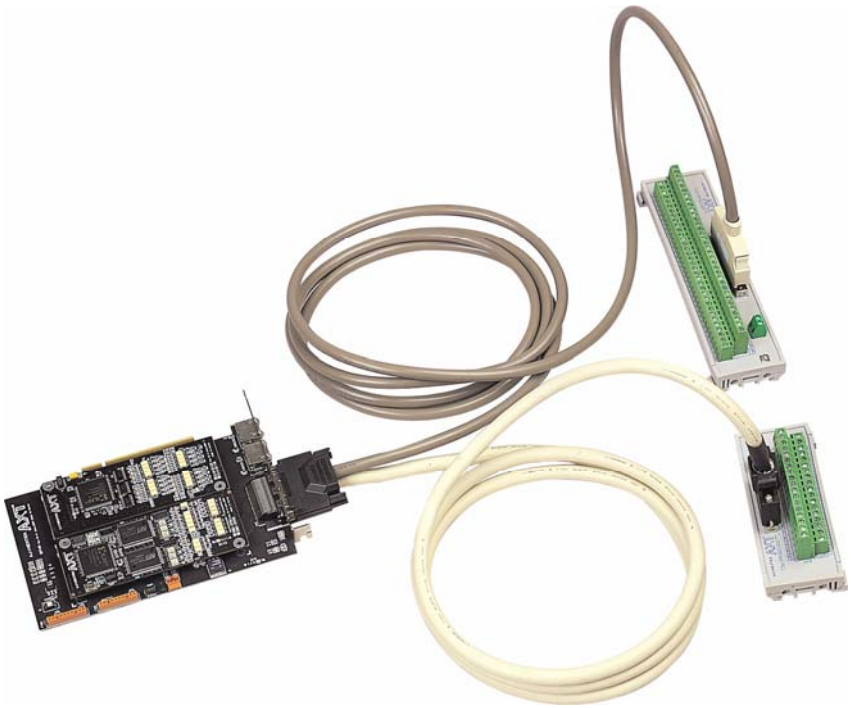
Module Name	Terminal Block	Cable
SMC-2V01 SMC-2V02 SMC-2V31 SMC-4V51 SMC-2V53 SMC-2V54	T68-PR	C68(0.8-1.27)
SMC-1V01 SMC-1V02	T36-PR	C68/36TS
SIO-DI32 SIO-DI16	APC-EI36 or T36-PR	C68/36TS
SIO-DO32P SIO-DO16T	APC-EO36 or T36-PR	C68/36TS
SIO-DB32P SIO-DB16T	APC-EB36 or T36-PR	C68/36TS
SIO-AI4R SIO-AO4R	T36-PR	C68/36TS

T68-PR terminal block can be applied to any kind of AnyPack series. However, SMC-1V01 and SMC-1V02, the 1 axis control module with a few signal line (less than 36 pins), AnyDIO, the digital I/O module and AnyAIO series, the analog I/O module are designed so that all signal lines to be inserted within 36 pins. The private terminal block of extended current drive capacity (500mA) can be used on the output stage using power transistor when big current output is required.

5.4.Device Connection



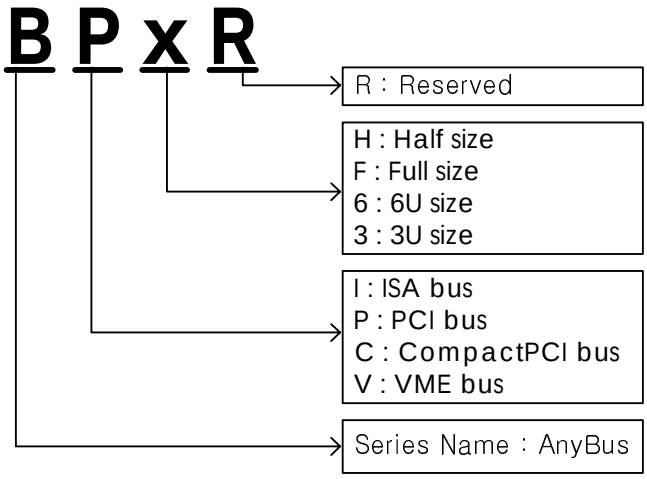
Picture 12. Connection between BPFR Board and the Peripheral (Terminal Board)



Picture 13. Connection between BPHR Board and the Peripheral (Terminal Board)

Picture 12 and picture 13 show the connections between the real BPFR, BPHR boards and terminal blocks. Picture 12 shows that BPFR's SUB1, SUB2, SUB3, SUB4 are loaded with SIO-DO32P (32ch digital output module), SMC-2V53 (2-axis servo module-PMD Chipset), SIO-DI32 (32ch digital input module), SMC-2V01 (2-axis servo module-CAMC Chipset), respectively, and connected with the terminal blocks of APC-EO36, APC-EI36, T-68PR by using C68-36 and C68(0.8-1.27). And picture 13 shows that BPHR's SUB1 and SUB2 are loaded with SMC-2V01 and SIO-DI32, respectively, and they are connected with each other by using terminal blocks and cables.

6. Ordering Information (AnyBus Family)



Picture 14. The Way of Reading AnyBus Series Model Name

Table 14. AnyBus Product Group

Model Name	Description
BIFR	ISA Full size baseboard(4 SUB slots)
BIHR	ISA Half size baseboard(2 SUB slots)
BPFR	PCI Full size baseboard (4 SUB slots)
BPHR	PCI Full size baseboard (2 SUB slots)
BV6R	VME 6U baseboard (4 SUB slots)
BV3R	VME 3U baseboard (2 SUB slots)
BC6R	CompactPCI 6U baseboard (4 SUB slots)
BC3R	CompactPCI 3U baseboard (2 SUB slots)

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